

Textbooks

- 1) Signals and Systems; 2nd Edition
 - By Oppenheim, Willsky and Nawab (OWN)
- 2) Digital Signal Processing: Principles, Algorithms and Applications; 4th Edition
 - By John. G. Proakis and Dimitris Manolakis (PM)

Class	Topics	Book sections	Practice problems
Mar 05	Introduction to the course	(PM) 1.1.1	
Mar 09	Sampling Process	(PM) 1.2.2, 1.2.3, 1.3.1, 1.3.2, 1.4, 1.4.1, 1.4.2	(PM) 1.2, 1.5, 1.7, 1.8, 1.15
Mar 12	Basic Signals	(PM) 2.1, 2.1.1, 2.1.3	(PM) 2.1, 2.2 (except 'g' and 'h'), 2.3
Mar 16	Discrete time Fourier Transform (DTFT)	(PM) 4.2.3,	(PM) 4.9a, 4.9b, 4.9c, 4.9f, 4.9g, 4.10a, 4.10b
Mar 23	Discrete Fourier Transform (DFT)	(PM) 7.1.1, 7.1.2, 7.1.3	(PM) 7.23, 7.24, ex7.1.1, ex7.1.2, ex7.1.3,
Mar 26	Zero padding & Time-domain Aliasing	(PM) 7.1.1	
Mar 30	Quiz – 1		

Class	Topics	Book sections	Practice problems
Apr 02	Discrete time systems	(PM) 2.2.1, 2.2.3	(PM) 2.6, 2.7, 2.12
Apr 06	Linear convolution	(PM) 2.3.3	(PM) 2.16, 2.18, 2.19, 2.21
Apr 09	Correlation	(PM) 2.6.1	(PM) 2.61, 2.62, 2.63,
Apr 13	Circular Convolution	(PM) 7.2.2, 7.3.1	(PM) 7.2, 7.8
Apr 16	Filters Introduction	(PM) 5.4.1, 10.1.2	
Apr 20	Filter Design	(PM) 10.2.2, 10.2.3	(PM) 10.1, 10.2, 10.5
Apr 23	Summary and Q&A session	--	--